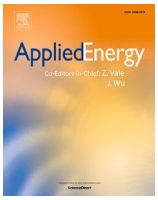
Applied Energy 392 (2025) 125990



Contents lists available at ScienceDirect

Applied Energy

journal homepage: www.elsevier.com/locate/apen





Multi-agent deep reinforcement learning based demand response and energy management for heavy industries with discrete manufacturing systems

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HIGHLIGHTS

- Multi-agent reinforcement learning framework is developed for demand response in energy-intensive industries.
- Centralized training with decentralized execution is proposed to optimize energy scheduling and coordination.
- Attention mechanism is integrated to enhance cooperation efficiency and improve policy stability.
- Proposed MARL-based strategy effectively balances energy cost reduction with production scheduling.
- Adaptive and scalable energy management approaches are demonstrated through case study.

ARTICLE INFO

Keywords:
Multi-agent reinforcement learning
Deep reinforcement learning
Demand response
Production scheduling
Energy management
Heavy industries

ABSTRACT

Energy-centric decarbonization of heavy industries, such as steel and cement, necessitates their participation in integrating Renewable Energy Sources (RES) and effective Demand Response (DR) programs. This situation has created the opportunities to research control algorithms in diverse DR scenarios. Further, the industrial sector's unique challenges, including the diversity of operations and the need for uninterrupted production, bring unique challenges in designing and implementing control algorithms. Reinforcement learning (RL) methods are practical solutions to the unique challenges faced by the industrial sector. Nevertheless, research in RL for industrial demand response has not yet achieved the level of standardization seen in other areas of RL research, hindering broader progress. To propel the research progress, we propose a multi-agent reinforcement learning (MARL)-based energy management system designed to optimize energy consumption in energy-intensive industrial settings by leveraging dynamic pricing DR schemes. The study highlights the creation of a MARL environment and addresses these challenges by designing a general framework that allows researchers to replicate and implement MARL environments for industrial sectors. The proposed framework incorporates a Partially Observable Markov Decision Process (POMDP) to model energy consumption and production processes while introducing buffer storage constraints and a flexible reward function that balances production efficiency and cost reduction. The paper evaluates the framework through experimental validation within a steel powder manufacturing facility. The experimental results validate our framework and also demonstrate the effectiveness of the MARL-based energy management system.

1. Introduction

The growing concerns over environmental sustainability, energy security, and socio-economic challenges have accelerated global efforts to integrate Renewable Energy Sources (RES) into power grids. The shift of focus towards RES has accelerated worldwide, with the adoption of policies and technologies supporting RES utilization across power systems. The transition has demanded sophisticated demand-side management

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https://doi.org/10.1016/j.apenergy.2025.125990
Received 12 February 2025; Received in revised form 10 April 2025; Accepted 24 April 2025
Available online 30 April 2025
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techniques, where Demand Response (DR) programs play a critical role in balancing supply and demand.

DR allows dynamic electricity consumption management and emerges as a pivotal tool in stabilizing power grids and optimizing energy use in line with generation and demand fluctuations. DR programs are generally categorized into price-based and incentive-based schemes. Oh et al. (2023) thoroughly discuss the dynamics of these schemes, with price-based DR allowing consumers to adjust consumption according to real-time electricity prices, while incentive-based DR offers rewards for reducing demand during peak periods [1]. Further, coordinating DR with renewable energy systems allows significant cost savings, enhanced energy efficiency, and environmental sustainability. Additionally, DR can be effectively integrated with energy storage systems to provide ancillary services, further supporting the integration of renewable energy into the power grid. This integration is essential for achieving a more flexible and resilient energy system, facilitating a transition to a sustainable future.

The research by Ye et al. (2024) further illustrates how DR functions as a crucial bridge in integrating RES [2]. Their work shows how DR scheduling of Hybrid energy systems (HESs) enables the integration of multiple energy sources, typically including renewable energy (such as solar, wind, and water powers) and traditional energy (such as coal, oil, and natural gas). Similarly, Oh et al. (2023) also present frameworks that merge energy storage with demand response, supporting global decarbonization initiatives [1].

Another aspect of DR research opportunities is the emphasis on optimization techniques and intelligent control systems. The review by Vázquez-Canteli (2019) highlights Reinforcement Learning (RL) as an adaptable approach to model reducing peak demand and managing energy loads, providing a comprehensive overview of RL-based DR as a potential solution for sustainable energy management [3]. Their work underscores how DR's efficacy is closely linked to the implementation of advanced metering infrastructure, automated control systems, and optimization algorithms for sector-specific energy management. Their findings stress that while DR in industrial sectors has yet to reach the level of residential or commercial applications, there is significant potential for Multi-Agent Reinforcement Learning (MARL) to streamline energy management in heavy industry.

Several factors contribute to the limited adoption of DR in industrial sectors. Industrial energy management must rely on diverse, energy-intensive processes with distinct operational needs. Furthermore, the diversity in energy usage patterns across various processes makes implementing a unified DR strategy complex. Lu et al. (2020) discuss the energy management challenges within discrete manufacturing systems, mainly focusing on the necessity for uninterrupted production, which constrains DR [4]. Zhu et al. (2022) expand on these insights, examining the energy management practices in multi-energy industrial parks and underscoring the importance of customized DR strategies that address these production-specific constraints [5].

Through the literature review, we highlight two significant gaps. First, a gap remains in exploring the application of MARL methods to heavy industries. Second, it relates to the first gap, as it stems from the fact that the standardization of RL research in the energy sector is predominantly focused on residential and commercial applications. RL research has made significant strides, primarily driven by the open-source community, which has developed standardized environments for testing and benchmarking RL algorithms. For instance, Moriyama et al. (2018) have introduced an OpenAI Gym interface for EnergyPlus, which serves as a testbed for RL in controlling HVAC systems in data centers [6]. Similarly, Puy-Velde et al. (2024) introduce HomeLabGym, which allows researchers to prototype, deploy, and analyze HEMS controllers within the controlled test environment of a real-world house, providing access to all its available sensors and smart appliances [7].

We understand that the gaps identified above can only be addressed by fostering research in industrial DR tailored specifically to heavy industries. To address these gaps in industrial DR, this study aims to develop a novel MARL framework for optimizing energy management in energy-intensive manufacturing processes. Specifically, we focus on the steel powder manufacturing (SPM) sector. Our method highlights that optimal energy scheduling requires the synergy among energy sources, storage, and loads, which necessitates the full cooperation of multiple energy devices. In this context, the MARL setting introduces each equipment load, storage, and source as an independent agent that can perceive the environment and adjust its energy policies [5]. The coordination among these individual agents can be fostered by centralized training and decentralized execution, i.e. a centrally trained value function helps agents learn individual policies. Further, we integrate an attention mechanism within the multi-agent framework to enhance coordination and maintain stationarity among agents. The critic is trained centrally with an attention mechanism, allowing it to prioritize relevant shared information. Meanwhile, individual agents operate based on local observations. This approach has demonstrated strong performance in the developed cooperative environments and previous studies [5,8]. Thus, the main contributions of this paper are summarized as follows:

- We extend existing research into a multi-agent reinforcement learning (MARL) setting, promoting further research efforts in DR for energy-intensive industries. This work highlights the potential of MARL in optimizing industrial energy management and addressing sector-specific challenges.
- To address the coordination overhead among individual agents, we propose a framework that utilizes centralized training while enabling decentralized execution. The central unit assists in formulating joint energy scheduling policies to optimize overall system performance.
- To enhance policy stability and improve cooperation efficiency among energy devices, we introduce an attention mechanism to the training framework. This mechanism allows agents to focus on critical energy-related information, such as time-varying demand and electricity prices, rather than processing all available data indiscriminately. By refining information selection, this approach improves exploration efficiency and enhances overall system performance.

Section 2 consolidates insights from previous studies, identifies gaps, and highlights novelty in our approach. Section 3 introduces the MARL framework, which uses a Partially Observable Markov Decision Process (POMDP) to model industrial environment settings. The framework incorporates equipment classification into load categories, formulating the optimization objective to achieve energy cost reductions without compromising production goals. Furthermore, we discuss the multi-agent proximal policy optimization (MAPPO) algorithm and the adaptation of the attention mechanism. In Section 4, we first discuss the steel powder manufacturing (SPM) process equipment and their operating parameters used in the MARL setting. Following this, we discuss in detail the experimental setup used in Python, which includes details on function implementations and open-sourced library dependencies. We utilize the use case study as the SPM to demonstrate the framework's effectiveness. Finally, we provide conclusions, discussions, and future directions in Section 5.

2. Literature review

Implementing DR strategies in energy-intensive industries has garnered significant research interest due to its potential to optimize energy costs, enhance grid stability, and facilitate the integration of renewable energy sources. Despite its demonstrated benefits in residential and commercial settings, adopting DR in industrial environments remains a complex challenge due to the diverse operational constraints, process dependencies, and production-critical requirements inherent in manufacturing systems. This section synthesizes key findings from existing research, highlighting how various optimization techniques, including mixed-integer linear programming (MILP), evolutionary algorithms, and RL, have been applied to improve energy management in industrial facilities. A particular focus is placed on the growing role of MARL as

Table 1
Related works on DR implementation.

Ref	Industry	Load types	Optimization method	Objectives
[10]	Steel manufacturing	Shiftable	MILP	Net energy cost minimization RES integration
[11]	Steel manufacturing	Shiftable	MILP	Net energy cost minimization
[12]	Steel manufacturing	Shiftable	MILP	Energy cost minimization
[13]	Steel manufacturing	Shiftable	MILP	Energy cost minimization Over-consumption penalty Under-consumption penalty
[16]	Steel Powder manufacturing	Shiftable Controllable Non-interruptible Energy storage	MILP	Energy cost minimization
[17]	Steel Powder manufacturing	Shiftable Controllable Non-interruptible	MILP	Energy cost minimization and forecast error minimization
[18]	Non-ferrous metal manufacturing	Shiftable	LP	Energy cost minimization
[19]	Automotive tire manufacturing	Non-interruptible Shiftable	MILP	Energy cost minimization RES integration
[20]	Metalwork industry	Shiftable	NSGA-II	Energy cost minimization Makespan minimization
[21]	Metal structure manufacturing	Shiftable	CG	Energy cost minimization
[23]	Automotive assembly	Shiftable	DMARL	Energy cost minimization
[22]	Cement manufacturing	Shiftable Energy storage	MILP	Daily profit maximization
[24]	Cement manufacturing	Shiftable Controllable	DRL	Energy cost minimization Resource storage monitoring
[25]	Steel Powder manufacturing	Shiftable Controllable Non-interruptible	DRL	Energy cost minimization with production constraints
[1]	Diverse industries	Energy storage	DRL	Energy cost minimization
[4]	Lithion-ion battery manufacturing	Shiftable	DMARL	Energy cost minimization
[5]	Diverse industries	Energy storage	DMARL	Net energy cost minimization RES integration
This work	Steel powder manufacturing	Shiftable Controllable Non-interruptible Energy storage	DMARL	Energy cost minimization Production maximization

an emerging solution to address the dynamic and cooperative nature of industrial energy scheduling. This review identifies gaps by analyzing prior studies across different industrial sectors and highlights the necessity for intelligent, adaptive DR frameworks tailored to energy-intensive manufacturing processes. These insights provide the foundation for our proposed MARL-based approach, which aims to enhance energy efficiency while maintaining operational feasibility in complex industrial environments.

2.1. Demand response-based energy management in energy-intensive industrial facilities

Table 1 provides an overview of various studies that have explored DR applications across different energy-intensive industries. The key parameters compared include the type of industry, the load types considered, the optimization methods used, and the optimization targets to be achieved. The studies span diverse sectors such as steel manufacturing, cement production, battery manufacturing, and more, reflecting the broad application of DR strategies. This subsection compares and contrasts these works, highlighting trends, limitations, and contributions.

Steel manufacturing is widely recognized as one of the most complex industrial sectors for implementing scheduling strategies [9]. The studies [10–15] focus on steel manufacturing. Studies [11,12] have analyzed the same steel manufacturing process, dividing it into four stages: electric arc furnace (EAF), argon oxygen decarburization units (AOD), ladle furnaces (LF), and continuous casters (CQ). While Zhang et al. [12] focuses only on optimizing electricity costs based on DR schemes, Zhang et al. (2015) use electric arc furnaces to offer spinning reserve services to the electricity markets [11]. Both studies define the optimization as a MILP. Both solutions employ the branch-and-bound (B&B) method, which stands out for its ability to systematically explore and prune the solution space, ensuring optimality while efficiently addressing the combinatorial nature of scheduling in steel manufacturing. In contrast, Pan et al. [13]and Cao et al. [10] investigate the steelmaking, refining, and continuous casting (SRCC) process. Pan et al. [13] specifically addresses a scheduling problem to reduce electricity costs and penalties for deviations from contracted loads.

While Cao et al. [10] considers real-time and demand charge tariffs in conjunction with self-generation. Both studies leverage a MILP model and utilize different variations of the strength Pareto evolutionary algorithm (SPEA) to solve the optimization tasks. Pan et al. [13] employs an enhanced SPEA2 algorithm (AHSPEA2) to optimize load tracking in SRCC processes. In their approach, the Pareto front represents the “best possible trade-offs” between competing objectives of energy cost minimization and penalties for over- and under-consumption. The Pareto front is a critical aspect of this approach, making it unique in explicitly addressing load tracking. Their computational results demonstrate the method’s effectiveness in closely following pre-contracted load curves, minimizing deviations, and reducing electricity costs. Similarly, Cao et al. [10] extends the application of SPEA2 to a steel plant equipped with self-generation under real-time electricity pricing. This study highlights the complexity of addressing centralized load shifting, which can lead to excessive load shifting during certain periods, necessitating the inclusion of demand charge pricing. The authors tailor the SPEA2 algorithm to include effective encoding mechanisms, conflict-resolution-based crossover, and mutation operators, and efficient decoding strategies that balance exploration and exploitation.

Two studies, [16,17], focus on the same steel powder manufacturing. They also share everyday use cases involving the same system organization and equipment. These studies consider various load types, including

shiftable, controllable, and non-interruptible loads, and aim to optimize energy management. Both [16,17] utilize MILP for optimization. While Huang et al. [16] focuses solely on energy cost minimization with the added consideration of energy storage systems, Lu et al. [17] incorporates day-ahead price forecast error minimization as an additional objective. These distinctions highlight the varying approaches that have addressed similar industrial challenges within the same operational context.

Different variants of linear optimization methods are used in various other industrial sectors. The case study presented by [18] is of a non-ferrous metal factory where the manufacturing process is modeled using the state task network (STN) framework. The work considers the production process to incorporate a melting electric furnace, a conversion furnace, electrolytic refining, and a fire refining furnace. Another study [19] also models industrial processes using the STN method. They include thermostatically controlled loads (TCLs) such as heating, ventilation, and air conditioning (HVAC) systems with chilled water storage (CWS), renewable energy generation from photovoltaic (PV) arrays, and electric vehicles (EVs). These components are modeled individually, and their operations are then optimized under time-of-use (TOU) pricing schemes. Factors influencing multiple components simultaneously, such as the number of workers, are also considered. The optimization problem is formulated as a MILP model. A general tire manufacturing facility serves as the case study for this approach.

Similarly, in [20], the authors explore various methods to create flexibility in industrial production, including changes in production sequence, load-shifting through rescheduling production pauses, and automatic adjustments of machine energy modes for additional energy savings. Specifically, an energy-adaptive production scheduling algorithm is introduced, tailored for job shops in the metalworking industry. This algorithm addresses two key objectives: it minimizes makespan and optimizes energy costs and associated greenhouse gas emissions through a time-of-use-based pricing scheme. To solve this multi-objective problem, the non-dominated sorting genetic algorithm-II (NSGA-II) is employed. Through Pareto optimization, the algorithm allows users to weigh energy costs and makespans according to their priorities.

The framework in [21] introduces an approach to facilitate and assess the integration of renewable energy into manufacturing systems by developing an optimization model for production scheduling that aligns with onsite renewable energy supply, supported by energy storage systems and the power grid as backups. Their work provides a case study in metal structure manufacturing. The model considers a multi-process production scheme alongside demand-side management policies, such as Time-and-Level-of-Use (TLU) and power consumption reduction requests. To account for the inherent uncertainties of renewable energy, a two-stage robust optimization model is formulated to optimize production scheduling under the worst-case scenario of renewable generation. A nested Column-and-Constraint Generation (CCG) algorithm is employed to solve this optimization problem. In [22], the authors highlight the critical role of industrial loads, such as cement plants, in enabling more flexible power system operations. The work divides the process into two stages: in the first, the machines crush raw material into finer particles; in the second, the furnace heats the resulting particles to a higher temperature for further processing in the following steps. The authors highlight that discrete power changes limit the ability of cement manufacturing to provide essential ancillary services like regulation and load following, which require continuous power adjustments. To address this limitation, the authors propose a novel approach that integrates an onsite energy storage system, allowing cement plants and similar industrial loads to offer these services by enabling more granular, continuous load adjustments.

The work [23] highlights the emergence of RL and presents considerable potential for real-time control in manufacturing. However, as production is the primary goal in manufacturing systems, an RL-based method must be effective, trustworthy, and explainable to ensure that production requirements are met. To address this need, their study proposes an explainable multi-agent deep RL approach, where a comprehensive manufacturing system model is used to decompose the overall energy management and production objectives into agent-level tasks. Each agent then identifies a safe subset of actions that align with the production requirements while simultaneously learning to optimize energy costs under DR. The decomposed multi-agent deep Q-network (DMADQN) method was applied to control an automotive assembly line using one year of DR electricity price data to validate its performance.

Ye et al. (2022) consider the cement manufacturing process [24]. They distinguish four key sub-processes in cement manufacturing: (1) Crushing (C), (2) Kiln Feed Preparation (KFP), (3) Clinker Production (CP), and (4) Finish Grinding (FG). They also consider storage silos placed between adjacent sub-processes, serving as intermediate product storage areas. The authors highlight that these silos not only help mitigate work efficiency mismatches between processes but also provide flexibility during power system disruptions. They adopt a DRL approach and define the Markov decision framework for the manufacturing process. They employ the proximal policy optimization (PPO) algorithm to train an agent in the manufacturing environment to minimize energy costs and over-and-under production penalties. The study [25] focuses on steel powder manufacturing, utilizing the same use case discussed in [16,17]. However, they differ significantly in their choice of optimization techniques and specific objectives. Huang et al. [25] employs deep reinforcement learning (DRL) to minimize energy costs while considering production constraints.

Several studies, such as [1,5,22], adopt a generalized framework applicable to diverse industries. Study [1] introduces a multi-agent framework for Energy Storage Systems (ESS) to participate in both price-based and incentive-based DR programs using RL on the demand side. While industrial and commercial customers are primarily subject to time-of-use (TOU) tariffs, the proposed framework specifically targets industrial users, allowing them to achieve additional profits by participating in incentive-based DR programs and managing their load more effectively. Industrial customers can shift their electricity consumption from high-price to low-price hours for arbitrage benefits, and they can also earn incentives by fulfilling dispatch signals from the system operator using ESS. To model ESS operation, the study applies a Markov decision process (MDP) in conjunction with MARL, enabling customers to make optimal decisions based on the current environmental conditions. The experimental results show that the proposed framework allows near-optimal decision-making for ESS's multiple uses using actual industrial load profiles and DR compensation rules in South Korea for training.

Similarly, in [5], the authors discuss the challenges of energy scheduling in an industrial park equipped with multi-energy devices, including generation, storage, and conversion systems. By treating the energy scheduling problem as a partially observable MDP, the framework achieves cooperative energy scheduling across distributed devices while reducing the impact of stochastic renewable energy and demand fluctuations. Both studies [1,5] highlight the synergy among energy sources, storage, and loads is crucial and ensures efficient collaboration through a decentralized execution framework, supported by centralized training. Lu et al. (2020) propose a general MADRL-based demand response scheme tailored for energy management in discrete manufacturing systems [4]. The manufacturing system is modeled as a partially observable Markov game, where each machine acts as an agent aiming to optimize its operation based on local observations. The work adopts the multi-agent deep deterministic policy gradient (MADDPG) algorithm to facilitate machine coordination to achieve an optimal energy usage schedule. The presented case study on a lithium-ion battery assembly manufacturing system illustrates the potential of the proposed framework. Furthermore, the paper compares the performance of the MADRL approach with traditional mathematical modeling methods, demonstrating its superior adaptability and cost-saving potential under dynamic conditions.

2.2. Reinforcement learning for energy management

Various energy management strategies have been developed for the smart grid ecosystem to enhance energy efficiency and cost savings, particularly by incorporating demand response mechanisms. These strategies can be broadly divided into model-based and model-free approaches. When demand response is framed as a discrete decision problem, subject to linear constraints and a linear objective function, it can be addressed using techniques like mixed-integer linear programming [26,27]. The system dynamics derived from the models can be used with model-predictive control approaches to formulate an optimization problem over a finite time horizon, allowing decisions on load adjustments. Optimization techniques such as the Strength Pareto Evolutionary Algorithm (SPEA) and Branch-and-Bound (B&B) have been widely utilized to enhance optimization in energy management scenarios, as seen in Section 2.1. SPEA provides a practical multi-objective optimization framework that balances trade-offs between energy cost minimization and other operational constraints. At the same time, B&B systematically explores the solution space by pruning suboptimal branches, ensuring computational efficiency. These methods, discussed in detail in the previous section, further strengthen decision-making in energy scheduling by offering robust solutions to complex industrial energy management problems. Furthermore, methods like stochastic programming [28] and robust optimization [29] account for uncertainties, such as fluctuating renewable energy output and variable electricity prices.

Model-based approaches rely on detailed models grounded in theoretical principles and often involve simplifications that may not fully align with real-world conditions [30]. High development effort, reliance on complex mathematical expressions, limited generalization across scenarios, extended computational times, and the absence of globally optimal solutions are some notable drawbacks associated with linear models and traditional optimization techniques [31]. In contrast, model-free techniques like RL can overcome these limitations, demonstrating high performance in demand-side management [3]. Further, RL mitigates challenges associated with high-dimensional data through function approximations [32]. Deep reinforcement learning (DRL) leverages neural networks for function approximation, enabling it to capture complex nonlinear patterns in data [33,34]. The adaptability of DRL allows the implementation of policy-based methods, particularly on-policy algorithms, and it has gained significant traction. These methods directly optimize control policies using neural networks without explicitly estimating value functions [35].

Addressing the limitations of traditional approaches, RL has emerged as a powerful tool for managing complex, dynamic systems. RL's ability to learn optimal policies through interaction with the environment makes it well-suited for industrial DR applications involving high-dimensional and variable operational constraints. Jain et al. (2024) demonstrate DRL's potential in optimizing power system operations, especially in managing complex scheduling and load-balancing tasks under constraints like uninterrupted production [36]. Their findings highlight RL as a key enabler for future industrial decision-making systems. Most DRL approaches for DR typically employ a centralized controller strategy where a single agent controls all energy management needs. Such centralized controllers use globally available information to compute actions that maximize a value function. However, such single-agent approaches may exhibit limitations like poor sample complexity and exponential growth of computational requirements as the demand response problem size increases [30].

Building on the principles of a partially observable Markov game (POMG), MARL extends DRL's applicability to scenarios where multiple agents must operate and interact within a shared environment. MARL is particularly relevant for industrial DR, where various energy systems, devices, or participants must coordinate actions to optimize overall energy consumption [37]. Salazar et al. (2024) emphasize the effectiveness of both cooperative and competitive multi-agent frameworks in enabling efficient DR modeling. Cooperative agents share global information to achieve common objectives, such as minimizing overall energy costs, while competitive agents handle situations where energy resources are contested [37]. Aladdin et al. (2020) explore the use of MARL for efficient demand response, detailing how multi-agent cooperation can optimize energy distribution and resource balancing by enabling each agent to adapt to grid conditions and production constraints [38].

Jiang et al. (2004) explore optimal scheduling strategies for distributed hydrogen refueling stations (HRS) using MARL [39]. Their study highlights that independent refueling stations do not fulfill market requirements to participate as ancillary services in the markets to provide DR. To tackle this problem, they present a model based on MARL, where distributed HRSs are aggregated as a single DR portfolio. Kumari et al. (2024) explore how decentralized residential energy management based on MARL can be integrated with blockchain technology to achieve secure and reliable decentralized DR [40]. The MARL approach in [40] addresses varying energy loads, including non-interruptible, shiftable, and controllable categories. In their framework, multiple agents are assigned to monitor and manage the energy consumption of home appliances corresponding to these load types. The agents are tasked with tracking appliance-level energy usage and implementing strategies to reduce overall consumption based on the operational requirements of the appliances. MARL is increasingly used in energy management systems to handle challenging cooperative learning tasks [8].

The adaptability of MARL makes it a promising approach for addressing the growing complexity of industrial systems and has gained significant research interest. MARL's model-free learning capabilities make it ideal for dynamic industrial environments that traditional optimization methods struggle to address. Huang et al. (2022) highlight how MARL-based optimization schemes handle uncertainties in real-time, enhancing production yield and ensuring uninterrupted production [41]. Liu et al. (2023) highlight that urgent demands and unexpected interruptions often affect production, and multi-agent systems help respond quickly to these dynamic disturbances [42]. Wang et al. (2022) examine the scheduling of discrete equipment in two-stage batch machines and dynamic job scheduling [43]. The method significantly reduces total tardiness by using a MARL approach and integrating batch formation and scheduling through two cooperative agents, outperforming traditional heuristics and other deep-learning techniques.

2.3. Identified research gaps and proposed contributions

The review of DR-based energy management in industrial settings reveals significant gaps in existing methodologies, particularly in the application of MARL for DR in heavy industries. While MARL has demonstrated effectiveness in managing energy consumption and optimizing DR strategies in other domains, its application in energy-intensive industrial environments remains largely unexplored. This gap presents an opportunity to develop novel frameworks that address the unique challenges of industrial settings, where cooperation among agents is crucial to balancing production objectives with energy efficiency. The key research gaps identified are as follows:

1. **Limited exploration of MARL in heavy industries for DR optimization**: Despite the proven advantages of MARL in fostering adaptive and intelligent energy management, its application in heavy industries remains insufficiently studied. Industrial processes are highly interdependent, requiring coordinated decision-making to balance energy efficiency with production goals. Existing studies primarily focus on price-based mechanisms and load shifting without fully leveraging MARL's potential to enhance cooperation among distributed agents. Addressing this gap is critical, as achieving cooperative behavior can significantly reduce production bottlenecks while ensuring energy efficiency.

2. **Need for novel frameworks to promote cooperation in industrial energy management**: Despite its potential, developing a MADRL-based demand response framework presents several research challenges [8]. While extensively studied in residential applications, these challenges are equally relevant to industrial settings. The first major challenge is designing energy management policies that foster cooperation among agents to minimize overall energy demand. Inefficient policies can lead to unintended peak load shifts rather than achieving the desired load flattening. Although tailored MADRL techniques have improved load shaping performance in grid-responsive buildings [3,44], their specificity can hinder adaptability to diverse real-world industrial environments. Conversely, applying conventional DRL algorithms in a multi-agent setting may result in high peak loads due to uncoordinated decision-making [8]. Another critical issue is the inherent non-stationarity of multi-agent environments, where dynamic interactions and information sharing among agents can destabilize learning. Ensuring stable policy convergence while maintaining system-wide coordination remains a key challenge in industrial energy management.

To address these gaps, our research methodology aims to:

- Design cooperative POMG framework that facilitates agent collaboration in heavy industrial settings, ensuring energy efficiency without disrupting production processes.
- Introduce a multi-agent deep reinforcement learning (MADRL) framework that leverages centralized training to enhance cooperative policies for efficient demand load shaping.
- Introduce an attention mechanism for information sharing to build coordination among agents and improve the learning efficiency of agents.
- Evaluate the impact of MADRL-based frameworks on optimizing energy consumption, mitigating peak loads, and improving overall industrial efficiency.

The applicability and efficiency of the proposed attention-based MADRL approach are validated through a case study application on an SPM process. The approach builds upon the MDP frameworks introduced in [25], extending it to a multi-agent partially observable Markov game (POMG). The choice of application in the SPM process is to maintain research comparison and reproducibility. Another factor is that the use case incorporates shiftable, controllable, and non-interruptible loads, allowing us to evaluate the cooperation among the three equipment types. In addition to the load types presented [25], we also incorporate energy storage systems, providing a more realistic representation of industrial operations. These modifications aim to enhance the application of MARL to implement DR in industrial settings.

3. Methodology

Developing an effective DR strategy for energy-intensive industries requires a structured approach to understanding energy consumption patterns, optimizing resource usage, and coordinating multiple energy assets. This methodology section introduces a comprehensive framework for modeling industrial manufacturing systems at the factory level, incorporating energy demand classification, mathematical modeling of energy and resource flows, and integrating energy storage systems. A key component of this framework is the application of MARL, which enables decentralized decision-making while ensuring system-wide coordination. By extending the MDP framework to a multi-agent POMG, this study models energy consumption and production scheduling as a cooperative optimization problem. To address the inherent complexity and non-stationarity of multi-agent environments, we incorporate an attention-based mechanism into the centralized critic of the Multi-Agent Proximal Policy Optimization (MAPPO) algorithm. This approach enhances coordination efficiency by selectively prioritizing relevant information, improving policy stability, and facilitating adaptive decision-making. The methodology outlined in this section provides the foundation for implementing an intelligent, scalable, and energy-efficient DR framework tailored for industrial applications, ensuring an optimal balance between cost reduction and production efficiency.

3.1. Manufacturing system organization and load classification

Braga-Santos et al. (2022) described and presented the organization of manufacturing systems by division into various levels [45]. The classification given by Braga-Santos et al. (2022) in [45] is first presented by Duflou et al. (2012) in [46] as an effort to review works related to energy and resource efficiency in manufacturing systems. Based on these works, the manufacturing activities of an industry/factory can be categorized into five organizational levels, varying from individual devices (in which unit processes occur) to the supply chain level (which incorporates the manufacturing ecology as a whole). Braga-Santos et al. (2022) highlight that the application of DR schemes can also be organized in these five levels [45]:

1. **Unit process level:** This level focuses on individual machines performing specific tasks. Optimizing these units is critical for energy efficiency and production stability.
2. **Multi-machine level:** Production processes involving interconnected machines that can be organized in series, parallel, or using a flexible job shop scheduling model.
3. **Factory level:** Processes that integrate and organize all equipment. At this level, the equipment considered is not strictly restricted to equipment used for manufacturing. They could be power storage or generation loads, water purifiers, cooling technology (including HVAC systems), lighting, and heating used in the factory or company.
4. **Multi-factory level:** Different facilities whose spatial organization allows for energy consumption, energy flow, and the reuse of waste synergies.
5. **Supply chain level:** This level comprises the manufacturing ecology as a whole and includes various individual facilities and other infrastructures such as transportation and other logistical externalities.

In our study, we look at the manufacturing system at the factory level. The energy management model at the factory level, based on [47], involves a Factory Energy Management System (FEMS) that coordinates components to manage energy and information flow. Fig. 1 depicts an example of an energy management model and various elements at the factory level. A smart meter (SM) tracks electricity between utility (UPL) and factory power lines (FPL). At the same time, a gateway (G) handles Demand Response (DR) signals between the wide area network (WAN) and local area networks (LAN). FEMS, at the core, controls communications, schedules primary energy loads based on production targets, and manages various other types of equipment. At this level, a general classification of the factory loads is achieved by a framework that identifies energy demands in the production process. Braga-Santos et al. (2022) have established the equivalence between the different load classifications presented in various literature focusing on DR in industrial settings [45]. Their work finds similarities in the classes discussed in related works and presents five distinct load definitions:

1. **Non-interruptible loads:** Loads or equipment that cannot be interrupted and whose energy demand must always be met regardless of the price of electricity.
2. **Shiftable loads:** Loads or equipment whose operation can be interrupted and whose energy demand can be adjusted (ON/OFF) if necessary.
3. **Controllable loads:** Loads or equipment whose operation can be interrupted and scheduled, comprising various operating points and energy demands.

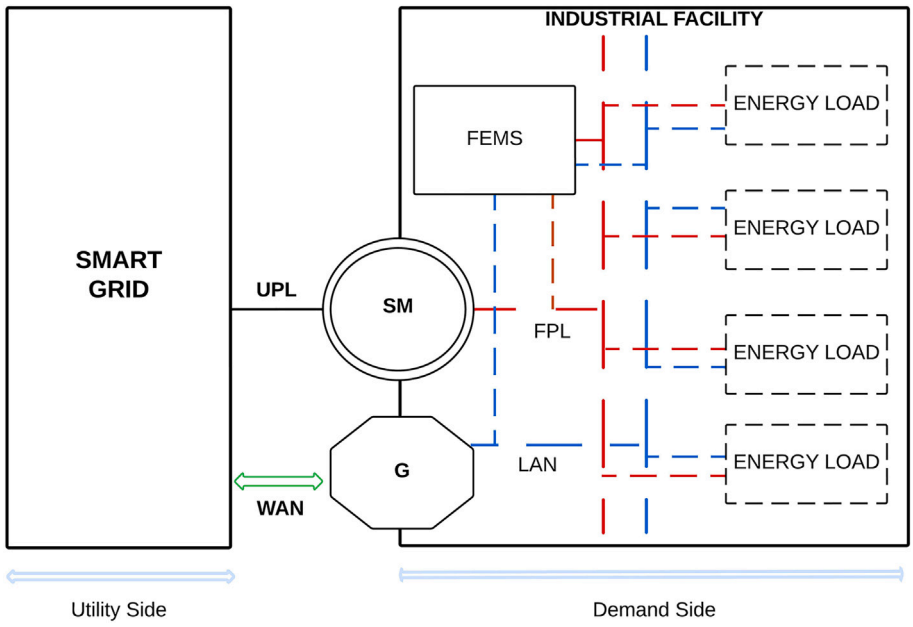


Fig. 1. Energy management model at factory level.

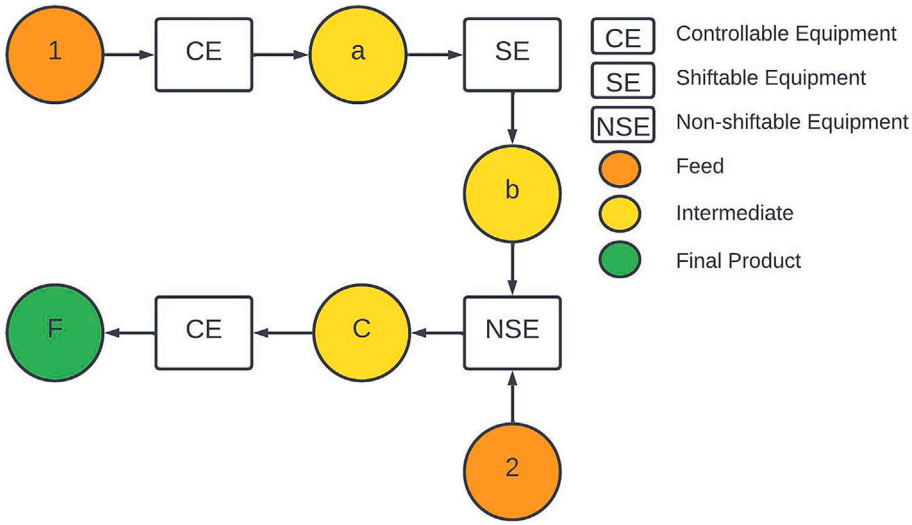


Fig. 2. An example of cooperation between different equipment types.

4. **Transferable loads:** Equipment that consumes a certain amount of electricity within a specified time interval. Outside this range, the equipment remains OFF, e.g. electric vehicles.
5. **Energy storage system:** Equipment such as batteries stores energy for later use.

Based on this classification of different equipment, a general mathematical model for energy demand and resource storage at the factory level can be proposed where DR schemes can be applied. An example of such a production process is shown in Fig. 2. In addition to categorizing the equipment based on energy demand and consumption levels, another aspect of such a model is the cooperation among the different equipment types. For example, in the process shown in Fig. 2, intermediate product "a" is consumed by a SE to produce another intermediate product "b". The intermediate material "a" itself is made by a CE. Therefore, the production of intermediate product "b" depends on the cooperation of two different equipment types. Thus, storage for intermediate resource "a" acts as a buffer that allows collaboration between the various pieces of equipment. Following the load classifications, in Section 3.2.1, we develop the mathematical model for energy demand and resource storage at the factory level.

3.2. Mathematical model

3.2.1. Production process energy demand modeling

We assume the factory processes can be modeled by discrete time steps $t(t \in T)$, and the EMS controls N equipment. During any random time step t, the operation point choice for any equipment $j(j \in [1, N])$ will be constrained by Eq. (1). Where, $\lambda_{i,t}^{j}$ is the binary indicator for operation point $i(i \in [1, I])$. Based on the operation point choice, the energy demand of a single equipment item i at stage t E_t^i is determined by Eq. (2), where the energy demand of the corresponding operation point is represented by e_t^i. The total energy demand of all equipment in an industrial process E_t^D is calculated using Eq. (3).

$$1 = \sum_{j=1}^{J} \lambda_{i,t}^{j} \tag{1}$$

$$E_t^i = \sum_{j=1}^{J} \lambda_{i,t}^{j} e_t^i \tag{2}$$

$$E_t^D = \sum_{i=1}^{N} E_t^i \tag{3}$$

3.2.2. Production resource balance model

Under the energy demand model defined earlier, the production balance model of the factory can be represented by Eq. (6). First, Eqs. (4) and (5) respectively establish the total production P_t^x and utilization U_t^x of a resource x at stage t, where $p_t^{j,x}$ is the production rate of resource x allowed by operation point j of equipment and $u_t^{j,x}$ the utilization rate allowed by operation point j of equipment that consumes x. At the end of stage t, as denoted by Eq. (6) R_t^x the residual of resource x, depends only on the amount stored during the previous stage R_{t-1}^x and recent production P_t^x and utilization U_t^x during the current stage t.

$$P_t^x = \sum_{i=1}^{N} \sum_{j=1}^{J} p_t^{j,x} \lambda_{i,t}^{j} \tag{4}$$

$$U_t^x = \sum_{i=1}^{N} \sum_{j=1}^{J} u_t^{j,x} \lambda_{i,t}^{j} \tag{5}$$

$$R_t^x = R_{t-1}^x + P_t^x - U_t^x \tag{6}$$

3.2.3. Energy storage system modeling

For each time step t, we define two binary indicators $\lambda_{c,t}^{ESS}$ and $\lambda_{d,t}^{ESS}$ that indicate the states of energy storage system i.e. charging and discharging respectively. If the ESS is charging at state t, $\lambda_{c,t}^{ESS} = 1$ and $\lambda_{d,t}^{ESS} = 0$. As the ESS cannot be charged and discharged simultaneously, the binary indications are constrained as follows

$$\lambda_{c,t}^{ESS} + \lambda_{d,t}^{ESS} \leq 1 \tag{7}$$

Let S_t^{ESS} denote the residual storage of energy in the ESS at time t. The evolution of S_t^{ESS} is dependent on the amount of energy charged $E_{c,t}^{ESS}$ or discharged $E_{d,t}^{ESS}$ during stage t. The residual energy in the ESS, S_t^{ESS}, can be computed as Eq. (8). We assume that the ESS has a charging efficiency of γ and a discharging efficiency of β. The parameters represent the energy conversion efficiencies of the ESS during charging and discharging processes, respectively. In practical applications, these values typically depend on the battery technology [48]. For instance, a lithium-ion ESS often has $\gamma = 0.95$ and $\beta = 0.92$, indicating that 5 % of energy is lost during charging and 8 % is lost during discharging. The formulation $(\gamma - \frac{1}{\beta}) \cdot E_t^{ESS}$ in Eq. (8) captures this asymmetry in energy loss. By including these factors, the model more accurately reflects the real-world degradation and round-trip efficiency of energy storage systems.

$$S_t^{ESS} = S_{t-1}^{ESS} + \lambda_{c,t}^{ESS} \cdot \gamma \cdot E_{c,t}^{ESS} - \lambda_{d,t}^{ESS} \cdot \frac{1}{\beta} \cdot E_{d,t}^{ESS} \tag{8}$$

The limited storage capacity F^{ESS} of the ESS introduces the following constraint on residual storage:

$$0 \leq S_t^{ESS} \leq F^{ESS} \tag{9}$$

3.2.4. Electricity cost

The net electricity demand of the facility E_t^{net} is determined using Eq. (10). In addition to the production process energy demand E_t^D, defined in Eq. (3), the calculation of the net electricity can incorporate various additional elements that influence the total electricity cost of the facility. The additional component can be expanded or modified to include multiple components such as energy storage devices, transferable loads, or other demand-side management strategies. We consider the ESS defined previously.

$$E_t^{net} = E_t^D - E_t^{ss} \tag{10}$$

Eq. (11) represents the energy cost function at stage t, where E_t^{net} is the net energy demand defined by Eq. (10) and HP_t is the hourly energy price.

$$C_t = E_t^{net} HP_t \tag{11}$$

3.2.5. DR objective

In most price-based DR scenarios, the consumption side seeks to minimize its energy cost, which depends on varying energy prices. With SE and CE in an industrial process, scheduling SE and CE during low pricing periods provides this DR potential. However, optimizing energy costs for industrial facilities can result in a trade-off with their production goals. The overall DR objective in an industrial setting, thus, is to minimize the total energy cost over all stages (Eq. (12)) while maximizing the total production over all stages (Eq. (13)).

$$\min \sum_{t=1}^{24} C_t \tag{12}$$

$$\max \sum_{t=1}^{24} \sum_{x} P_t^x \tag{13}$$

The objective functions in Eqs. (12) and (13) define two conflicting objectives. These objectives are conflicting, as improving one could degrade the other. To balance these objectives, a trade-off parameter α can be used to obtain a weighted sum of the two objectives, Eq. (14).

$$\max \sum_{t=1}^{24} \left(\alpha \sum_{x} P_t^x - (1-\alpha) C_t \right) \tag{14}$$

The weight parameter α varies between 0 and 1, controlling the trade-off between maximizing production and minimizing cost. The effect of the weight parameter α is crucial for understanding the interaction between the environment and the reward optimization process. By adjusting α, one can influence the balance between cost optimization and production optimization, ultimately impacting the agent's performance. Therefore, it is important to investigate how varying α affects the reward structure and the agent's behavior. An ablation study of this parameter is thus presented in Section 4.3.1 to analyze its influence on the overall system's performance.

3.3. MARL setting

In this section, we extend the Markov Decision Process (MDP) framework introduced in [25] to a multi-agent partially-observable Markov game (POMG). The multi-agent POMG is defined as a 4-tuple $(\mathbf{S}, \mathbf{O}, \mathbf{A}, \mathbf{R})$, where $\mathbf{S}$ represents the set of environment states. The set of observations, denoted by $\mathbf{O}$, defines the information each agent acquires about the system state. Each agent selects an action part of the set $\mathbf{A}$. Each agent then receives a reward that belongs to the set of rewards $\mathbf{R}$. Each agent $j (j \in [1, N])$ chooses actions given an observation guided by a policy $\mu_j : o_j \rightarrow a_j$, which maps local observations o_j to corresponding local actions a_j. Upon executing actions a_j, agents receive rewards r_j, and the global system state s transitions to a new state s' based on the environment dynamics.

3.3.1. System state formulation

The individual state observations of agent, $j (j \in [1, N])$, o_j constitute an internal state $S_j^{internal}$ and an external state $S_j^{external}$. At each timestep

t, the internal state $S_{j,t}^{internal}$ for equipment includes the residual of intermediate resources consumed and/or produced by the equipment. For example, if equipment agent j consumes resource a and outputs resource b, the internal state $S_{j,t}^{internal}$ would be the residual of input resource R_t^a and the residual of output resource R_t^b. If any equipment $j(j \in [1, N])$ does not require input resources (for example, a water cooler or ESS where residual is the stored energy), the internal state would only correspond to the residual of the output (i.e. the amount of cool water and the amount of stored energy).

The external state $S_j^{external}$ is defined by time-related factors such as the current hour of the day and the electricity prices provided by the smart grid (SG). Therefore, the system state S_t defined by the real-time electricity price HP_t, a time indicator of the current stage t, and the observation vector $\vec{o^j}$ would be defined as

$$S_t = t \times HP_t \times \vec{o^j} \tag{15}$$

3.3.2. System action formulation

The system action at any stage a_t is simply the set of operation states of each agent. For equipment agents in the production process, $a_{j,t}$ is merely the operation point. For agents such as the ESS, $a_{j,t}$ would be the choice of charging or discharging. If we consider $\vec{a}_{j,t}$ to be the vector of the operation point selected by each agent, Eq. (16) represents an example of the system action a_t, which is composed of all the operation point choices.

$$a_t = \vec{a}_{j,t} \tag{16}$$

3.3.3. Reward function

The reward function is formulated by Eq. (19) where α is the weight parameter for two objectives: cost optimization defined in Eq. (17) and production optimization defined in Eq. (18).

$$r_t^c = E_t^{net} HP_t \tag{17}$$

$$r_t^p = \sum_1^x -(\max(R_t^x - \text{capacity}_{\max}^x, 0) + \max(\text{capacity}_{\min}^x - R_t^x, 0)) \tag{18}$$

$$r_t = (1-\alpha) r_t^p - \alpha r_t^c \tag{19}$$

The production reward function r_t^p, rewards the agent for producing intermediate resources while maintaining storage constraints. Maximizing r_t^p ensures production efficiency. On the other hand, the cost reward function r_t^c penalizes the agent for energy consumption. From the observations of ablation study presented in Section 4.3.1, the value of α is set to 0.8. Since the agents are trained to optimize a global reward, the setting is thus a multi-agent cooperative learning. In this setting, each agent acts according to its policy μ_{θ_j} to optimize a shared reward $r_j = r$. The optimization problem is to find a policy that maximizes this expected shared return for all agents. This problem can be solved using the Multi-agent Proximal Policy Optimization (MAPPO) algorithm described in the following subsection.

3.4. Multi-agent proximal policy optimization (MAPPO)

MAPPO [49] is an extension of the Proximal Policy Optimization (PPO) algorithm [35] to multi-agent environments. Like PPO, MAPPO training consists of optimizing two networks: an actor network and a critic network. The actor–network maps the local state of an agent to optimal actions. The critic network evaluates the actions received from the actor network to improve the performance of the actor network. Unlike optimizing a single policy network in PPO, MAPPO extends to the decentralized actor and centralized critic framework. This framework is achieved by training the individual actor networks using only the local observation state of particular agents. In contrast, the critic networks use the states and actions of all agents in the system to provide feedback on each agent's policy. This training process is also known as the Centralized training with decentralized execution (CTDE) approach, where agents have access to global information while training but act based on local observations during execution. The CTDE execution addresses the non-stationarity of the environment from each agent's perspective in a fully decentralized training approach [50].

3.4.1. Algorithm

The details of the MAPPO implementation are presented in Algorithm 1. The learning process starts with initializing policy networks μ_{θ_j} for each agent j. The policy networks are parameterized by weights θ_j and are used to compute the local observations of agents. Similarly, the shared centralized value function $V_\phi(s)$, parameterized by ϕ, is initialized and will be used to estimate the global state s value. Using an initial state $s_j^t \in \mathbf{s}$ as input, an agent j takes an action $a_j \in \mathbf{a}$ given by:

$$a_j^t \sim \mu_{\theta_j}(a_j^t | s_j^t), \tag{20}$$

The actions computed by all the agents are applied to the environment, and each agent receives the shared reward $r^t \in \mathbf{r}$ calculated according to Eq. (19) and observes a new global state $s' \in \mathbf{s}'$. The experience $(\mathbf{s}, \mathbf{a}, \mathbf{r}, \mathbf{s'})$ is then stored for policy update in batch, and the initial global state is updated: $s \leftarrow s'$.

After every ζ number of episodes, for each agent j, the actor μ_j and critic $V_\phi(s)$ networks are updated using the current experiences stored. Let (s_i, a_i, r_i, s_i') be an experience for a transition $i, i \in n$. Each agent j computes the Generalized Advantage Estimation (GAE) value given by:

$$A_j^t = \sum_{k=0}^{\infty} (\gamma\lambda)^k \delta_j^{t+k} \tag{21}$$

where δ_j^t is the temporal difference (TD) error given by

$$\delta_j^t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t) \tag{22}$$

After computation of the advantage values, each agent j optimizes its policy loss given by,

Algorithm 1 MAPPO algorithm.

```
For each agent j, randomly initialize policy networks μ_{θ_j}
initialize a centralized value function V_φ
for episode = 1 to M do do
    Initialize environment and observe the initial global state s
    for timestep = 1 to T do do
        for each agent j = 1 to N in parallel do do
            Compute actions using Eq. (20)
        end for
        Execute actions a_i in the environment
        compute reward r, obtain new agent states s'_j and global state s'
        Store transition (s, a, r, s') in replay buffer D
        Update state: s ← s'
    end for
    if episode %ζ = 0 then then
        for each agent j = 1 to N in parallel do do
            Sample mini-batch of transitions from replay buffer D
            Compute advantage estimates Eq. (21)
            Update θ_j by optimizing loss given by Eq. (23) using Adam
            Update φ by optimizing loss given by Eq. (24) using Adam
        end for
    end if
end for
```

$$L^{\text{policy}}(\theta_j) = \mathbb{E}^t\left[\min\left(r^t(\theta_j)A_j^t, \text{clip}(r^t(\theta_j), 1-\epsilon, 1+\epsilon)A_j^t\right)\right] - \beta\mathbb{E}^t\left[\mathbb{H}(\mu_{\theta_j})\right] \tag{23}$$

where,

$$r_t(\theta_j) = \frac{\mu_{\theta_j}\left(a_j^t|s_j^t\right)}{\mu_{\theta_j^{\text{old}}}\left(a_j^t|s_j^t\right)}$$

and

$$\mathbb{H}\left(\mu_{\theta_j}\right) = -\sum_a \mu_{\theta_j}\left(a_j^t|s_j^t\right)\log\mu_{\theta_j}\left(a_j^t|s_j^t\right)$$

The clipping operation ensures the stability of the policy function by constraining the policy update. And the entropy term $\mathbb{H}(\mu_{\theta_j})$ penalizes deterministic policies by encouraging higher entropy (i.e. promoting exploration). Finally, the parameters of the centralized value function $V_\phi(s_t)$ are updated by minimizing its mean squared error, compared to the discounted return G_t:

$$L^{\text{value}}(\phi) = \mathbb{E}_t\left[\left(V_\phi(s_t) - G_t\right)^2\right] \tag{24}$$

where the discounted return G_t is computed as:

$$G_t = \sum_{k=0}^{\infty}\gamma^k r_{t+k} \tag{25}$$

Optimization algorithms such as stochastic gradient descent (SGD) and its variants, like Adam, can update the parameters of the policy and value networks based on the computed loss functions.

3.4.2. Attention-based value function estimation for critic

As discussed earlier, the ability to accurately evaluate the value of states and actions is crucial for effective decision-making in multi-agent reinforcement learning. In the MAPPO algorithm discussed in Section 3.4.1, the centralized critic plays a key role in evaluating the global state to provide feedback to decentralized actors. However, a challenge arises when agents need to coordinate efficiently with each other in a large-scale multi-agent setup. As the number of agents increases, the amount of information required for the critic to evaluate becomes more complex. This results in significant communication overhead, and not all information is relevant at every time step for each agent [5].

To address these challenges, studies like [5] and [8] propose incorporating an attention mechanism into the centralized critic, enhancing the efficiency and effectiveness of the value function estimation. The attention mechanism allows the critic to selectively focus on the most relevant information from the global state, filtering out unnecessary data and highlighting pertinent information, and also reduces the computational burden. This mechanism is particularly useful in dynamic environments where the non-stationarity introduced by multiple agents may lead to inefficiencies in coordination and cooperation.

In the MAPPO setup, described in Section 3.4.1, the centralized value function $V_\phi(s)$ takes the states and actions of all agents to estimate the value of the global state. However, using an attention mechanism, the centralized critic can be modified to focus on the most relevant contributions from each agent. At each timestep, instead of using the global states of the agents, the critic utilizes an attention-based approach to selectively weigh the contributions from each agent based on their relevance to the current decision. The network architecture of the attention-based critic is shown in Fig. 3

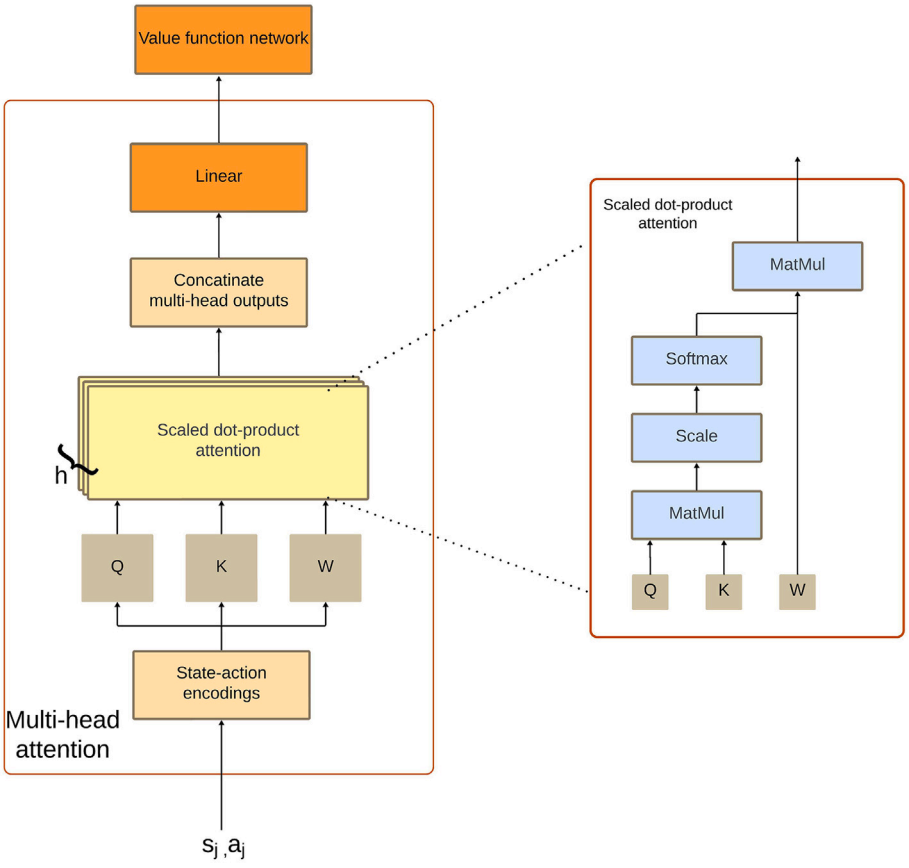


Fig. 3. Architecture utilizing multi-head attention.

The attention mechanism shown in Fig. 3 is also referred to as "Scaled dot-product attention" or "self-attention" and was first described by Vaswani et al. (2023) [51]. Scaled dot-product attention computes attention scores by comparing query, key, and value representations of embedded input tokens. The self-attention mechanism allows the model to weigh the importance of different elements in the sequence. As seen in Fig. 3, this mechanism is formalized using two steps. First, a softmax function is applied over the dot product of the query and key vectors, normalized by the square root of the dimension. The softmax function thus outputs the "attention weights", which are multiplied by the value vector to assign different levels of importance to various parts of the input and obtain filtered information. A multi-head attention framework can be developed by stacking multiple attention layers and combining them with feedforward networks. By doing so, the model can process and synthesize information in various ways, thereby greatly enhancing its capabilities.

To implement this attention-based framework for the centralized critic, we start by representing each agent's local state–action pair (s_j, a_j) using embeddings. The embedding layer transforms the state–action pair (s_j, a_j) into a latent feature space $Z_j = Embed(s_j, a_j)$. The attention mechanism is then applied. The attention weights are first calculated using Eq. (26)

$$a_{i,j} = Softmax\left(\frac{Q(z_j).K(z_j))}{\sqrt{d}}\right) \tag{26}$$

where $Q(z_j)$ and $K(z_j)$ are query and key transformations of the embedding, and d is the dimension of the query and key vectors. The computed attention weights are then used to aggregate the filtered information from all agents.

$$\tilde{z}_i = \sum_{j \neq i} a_{i,j} \mathrm{W}(z_i) \tag{27}$$

where $\mathrm{W}(z_i)$ is the value transformation of the state–action embedding. Finally, this aggregated information is passed through the value network to estimate the value function.

$$V_\phi(s_t) = f_\phi\left(\sum_{j=i}^{N} \tilde{z}_j\right) \tag{28}$$

The loss function for the value function remains the same as in Eq. (24) where the value function computation is now given by Eq. (28). With this new value function computation, the policy loss function for the decentralized actors remains the same as in standard MAPPO. The temporal difference (TD) advantage δ_j^t used in the advantage function A_j^t in Eq. (21) now is calculated using the new attention-weighted value received from the critic.

4. Numerical study and results

4.1. Steel powder manufacturing process

The SPM process described in [16,25] is shown in Fig. 4. The process is linear and divided into three phases according to intermediate products. The process requires three significant production steps: iron powder fabrication, steel powder formation, and steel powder refining. The first step of iron powder fabrication is the key component of the entire process, the atomizer transforms molten iron into a solidified droplet. When the molten iron falls into the atomizer, several high-pressure water jets form the liquid iron into metal droplets. Then, the slurry will be sieved in the specialized dehydrator, and the remaining solidified droplet will be heated in the dryer to evaporate any residual moisture.

The second step is steel powder making. The dry powder produced in the previous step must be homogenized in a crusher and classifier to ensure powder size uniformity. After that, the semifinished iron powder is poured into a magnetic separator to remove any remaining mixed slag. Then, the iron powder will be treated in a reduction furnace to reduce the oxide and carbon content and produce steel powder. The steel powder must be further refined in the third step. Therefore, the crusher and classifier are used to control the particle diameter. A blender may mix other materials into the powder to satisfy market demands [16].

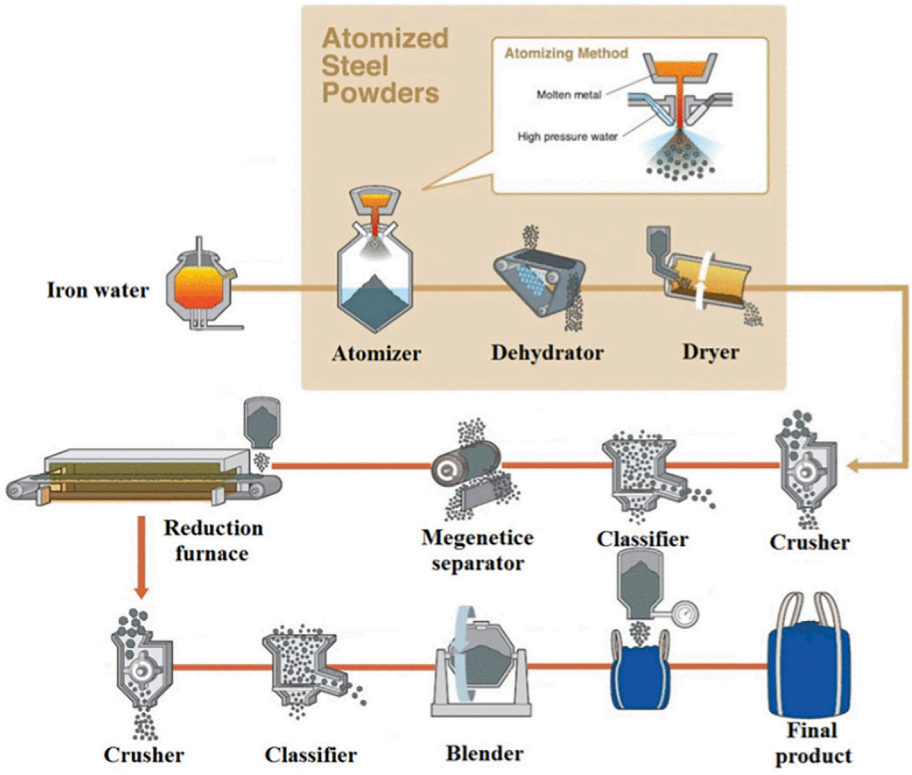


Fig. 4. Steel powder production process as described in studies [16,25]. Figure taken from [25].

Auxiliary equipment is essential to the primary manufacturing process. One is the water cooling tower, which generates cooled water. Cooled water is indispensable for atomization and as a coolant for the dryer and reduction furnace. Another piece of equipment is a nitrogen generator, which provides pure nitrogen to the reduction furnace to guarantee an oxygen-deprived atmosphere.

Table A.6 lists all equipment related to SPM. The equipment is classified according to the categories discussed in Section 3.1. The reduction furnace is an NSE with only one OP and is assumed to be operational regardless of the electricity price. Meanwhile, the atomizer, dehydrator, dryer, and magnetic separator are classified as SE because they only have two states: ON and OFF. The crusher, classifier, blender, WCT, and NG are classified as CE due to their multiple working states with different resource processing rates. The processing rates of equipment in the manufacturing process are shown in Table A.7. Similarly, Table A.8 and Table A.9 show relevant equipment's water processing and nitrogen processing capabilities. Tables A.10–A.12 provide all information on resource storage related to SPM. Specifically, Table A.10 shows the resource storage parameters in the primary process. Correspondingly, Tables A.11 and A.12 present detailed information on cool water storage and pure nitrogen storage, respectively. Finally, Table A.13 presents parameters for an ESS. The battery has a full capacity of 300 kWh. The charging and discharging efficiency is set to 90 %. The maximum charging and discharging rates were set to 50 kWh, i.e. the maximum energy that can be charged or discharged in 1 h. As discussed in Section 3.2.5, the primary objectives of the DR-based EMS scheme applied to the discussed SPM process are:

1. Minimize energy costs by dynamically adjusting SE and CE energy consumption based on time-varying electricity prices, coordination with ESS, and availability of RES.
2. Maintain production targets to ensure uninterrupted manufacturing operations.

4.2. Experimental setup

The experimental setup created for the MARL is represented in Fig. 5. The process simulation is achieved by creating a factory class using standard Python class definition. The factory class inherits Python object instances created for load types discussed in Section 4.1. The load instances are initialized with respective equipment parameters, and storage resource outlets are monitored. The factory class also has a method defined that uses actions selected by agents in the MARL environment to run the SPM process and return computed rewards and updated environment state. The Factory object instance created for the SPM process is passed into a $PettingZoo$ multi-agent environment class. $PettingZoo$ is akin to a multi-agent version of Gymnasium [52] and provides a standard API for multi-agent RL environments.

For training the multi-agent PPO discussed in Section 3.4.1, we leveraged the $Ray - RLlib$ framework. The Ray framework provides a universal API for building distributed applications [53]. $Ray - RLlib$ is an open-source library for reinforcement learning that leverages nested parallelism from the Ray framework to achieve state-of-the-art training performance for RL workloads [54]. The library offers a collection of reference algorithms and scalable abstractions for quickly composing new ones. $Tune$ is another library we leverage during training. Built on top of Ray, $Tune$ library allows scalable hyperparameter tuning in distributed reinforcement learning [55].

The $PettingZoo$ environment interacts with $Tune$, using a wrapper function provided by Ray is utilized. The $ParallelPettingZooEnv$ wrapper thus acts as an interface between the environment and RLlib's multi-agent API. We use PyTorch as the backend to build policy and value networks (representing policy functions for each agent and a central value function). Table 2 presents the network architecture details. The policy network does not utilize the attention mechanism. It comprises two fully connected hidden layers with 400 and 300 neurons,

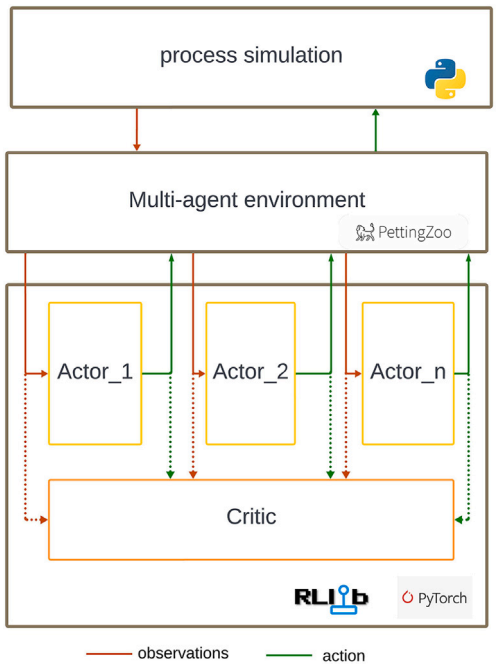


Fig. 5. Experimental setup SPM energy management using multi-agent reinforcement learning and MAPPO. The dotted lines represent shared action and observations passed to the central critic.

Table 2
Summary of actor and critic network parameters used in the numerical study.

Parameter	Actor network	Critic network
Use attention	No	Yes
Number of attention heads	–	4
Attention dimension	–	256
Attention head dimension	–	64
Attention MLP dimension	–	256
Activation function (Hidden Layers)	ReLU	Leaky ReLU
Final activation function	Tanh	Tanh
Hidden layer dimensions	(400, 300)	(400, 300)

respectively. These hidden layers are activated using the *ReLU* activation function. In contrast, the value network (critic) incorporates the attention mechanism, allowing it to process temporal dependencies in the data more effectively, as discussed in Section 3.4.2. The critic network has four attention heads, each with a dimension of 64, resulting in a total attention dimension of 256. Like the actor network, the hidden layers of the critic network consist of 400 and 300 neurons, with *Leaky ReLU* activations applied between the layers.

Table 3 presents the parameters used during training. The value network loss was assigned a coefficient of 0.2, with the clipping parameter set to 0.1. The entropy regularization coefficient β in Eq. (23) was set to 0.9 to encourage policy exploration without additional entropy scheduling. The training setup utilized a GAE factor (λ in Eq. (21)) of 0.8 to balance bias and variance in the advantage estimation. The discount factor (γ) was set to 0.9. KL-divergence-based policy updates were utilized to ensure stable learning, using a KL coefficient of 0.8 and a target divergence of 0.001. Learning was conducted over 100 epochs using a learning rate of 5×10^{-4}, with a batch size of 24. The training process utilized four environment runners, each with a single environment instance, running in parallel on one CPU per runner.

To verify the performance of the proposed algorithm, we compare it against several baseline methods, including the Proximal Policy Optimization (PPO) algorithm [35] and the Multi-Agent Proximal Policy Optimization (MAPPO) algorithm [49]. While PPO is a single-agent implementation, MAPPO and the proposed algorithm are designed for

Table 3
Summary of training setup parameters.

Parameter	Value
Value network loss coefficient	0.2
Clipping parameter	0.1
Entropy regularization coefficient (β)	0.9
GAE factor (λ)	0.8
Discount factor (γ)	0.9
KL coefficient	0.8
Target KL divergence	0.001
Learning rate	5×10^{-4}
Batch size	24
Number of epochs	100
Number of environment runners	4
CPU per runner	1

Table 4
Comparision of different methods.

Methods	Environment interaction
PPO	Single agent interacts with environment state.
MAPPO	Individual agents interact independently, coordinated by a centralized critic.
Proposed algorithm	Attention mechanism to prioritize important agents or states for decision-making.

multi-agent settings. MAPPO and the proposed algorithm utilize centralized critics and decentralized actors, with the key distinction that the proposed algorithm incorporates an attention mechanism in the critic. This attention mechanism allows the critic to selectively focus on relevant agents during learning, enhancing coordination and decision-making in multi-agent environments. A summary of the comparison between the different methods is presented in Table 4.

4.3. Numerical results

Fig. 6 demonstrates the system's convergence towards the optimal reward value for different methods. Initially, all methods operate with minimal understanding of the actions required to achieve high rewards, leading to a phase of random exploration. Over time, as the iterations progress, the methods accumulate experience by repeatedly interacting with the environment. The proposed algorithm, leveraging the attention mechanism in the critic network, achieves faster convergence compared to standard PPO and MAPPO. Specifically, the proposed algorithm shows stable convergence in approximately 200 training episodes, whereas MAPPO, which also employs a centralized critic but lacks the attention mechanism, although showing early convergence, does not exhibit stability. As a single-agent variant, PPO exhibits the slowest convergence and also achieves a convergence towards a lower episode reward.

Fig. 7 illustrates the energy demand of the factory across discrete time steps, with results from different methods. The figure highlights a clear pattern where machines increase electricity usage at low prices and reduce consumption during high-price periods. This behavior demonstrates load shifting. However, a closer inspection of the plot shows that the load-shifting behavior of the PPO algorithm is based on an inefficient policy that leads to unintended peak load shifts rather than achieving load flattening. This behavior is seen in the peaks around timesteps 1–3. Compared to this behavior of the PPO policy, both MAPPO and the proposed algorithm show the desired load shaping, avoiding unwanted peak load shifts. As discussed earlier in the literature review section, the MADRL-based techniques have demonstrated improved load-shaping performance.

Table 5 and Fig. 8 present the total costs across 24 time steps, achieved by each method. In Table 5, we also compare the "skill" of the multiagent models. The skill is computed as the % in cost reduction achieved by the multiagent model compared to the single agent PPO. The proposed algorithm achieves lower costs than PPO and MAPPO. Since the agent's objective was to achieve cost reductions and optimize production, we analyzed the impact of achieved cost reductions on environmental dynamics and production patterns.

To monitor resource storage effectively, a heat map can visually represent container utilization over time. The heat map will have containers on one axis and time on the other, with the color intensity representing each container's resource utilization percentage at each time step. The rate is calculated relative to the maximum capacity of each container, where 100 % indicates the container is full. This visualization

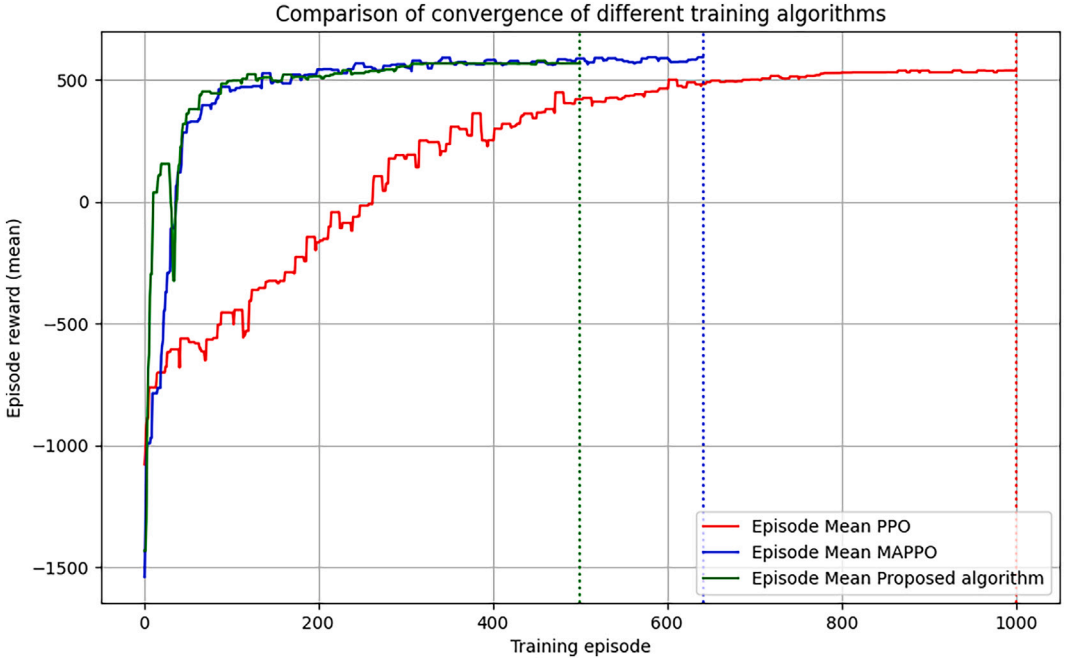


Fig. 6. Episode rewards convergence shown by different methods.

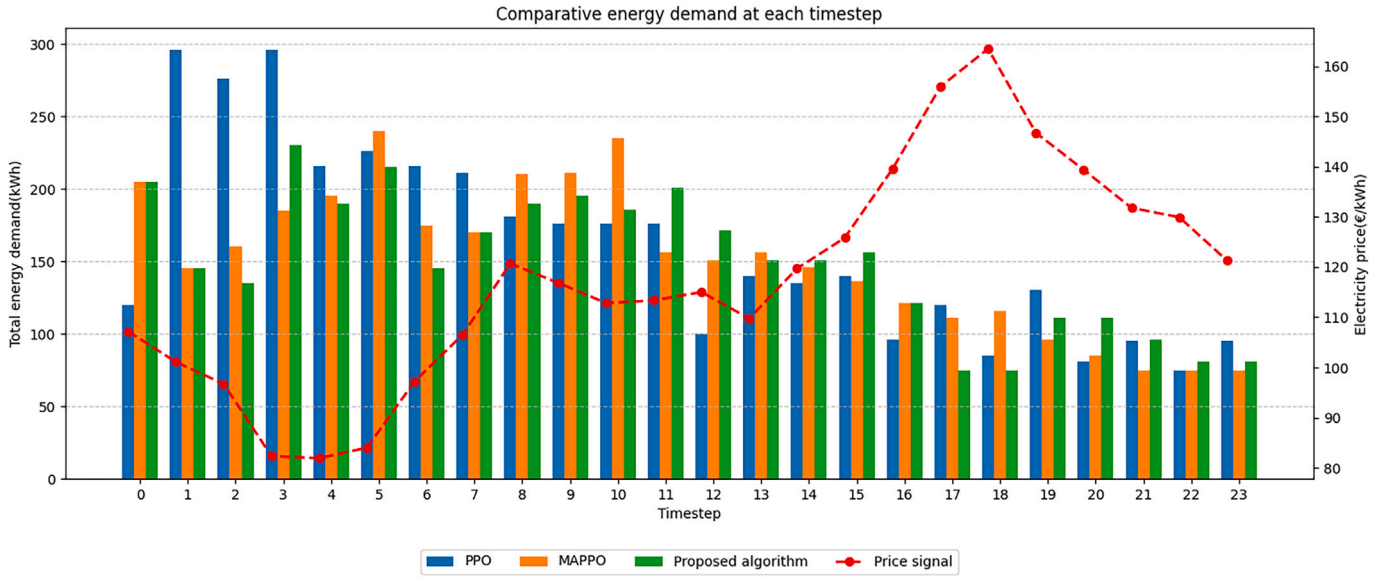


Fig. 7. Comparative energy demand across discrete time steps.

Table 5
Comparison of total costs across methods.

Methods	Total costs (€)	Skill
PPO	426.3	–
MAPPO	408.2	4.2 %
Proposed algorithm	401.2	5.8 %

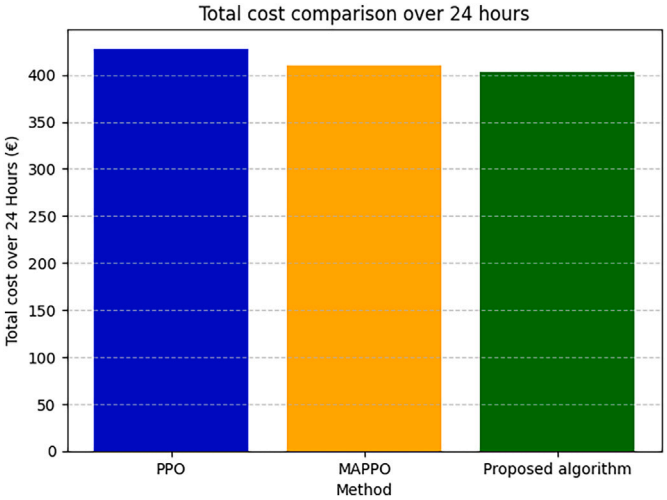


Fig. 8. Total cost of SPM process achieved by each method.

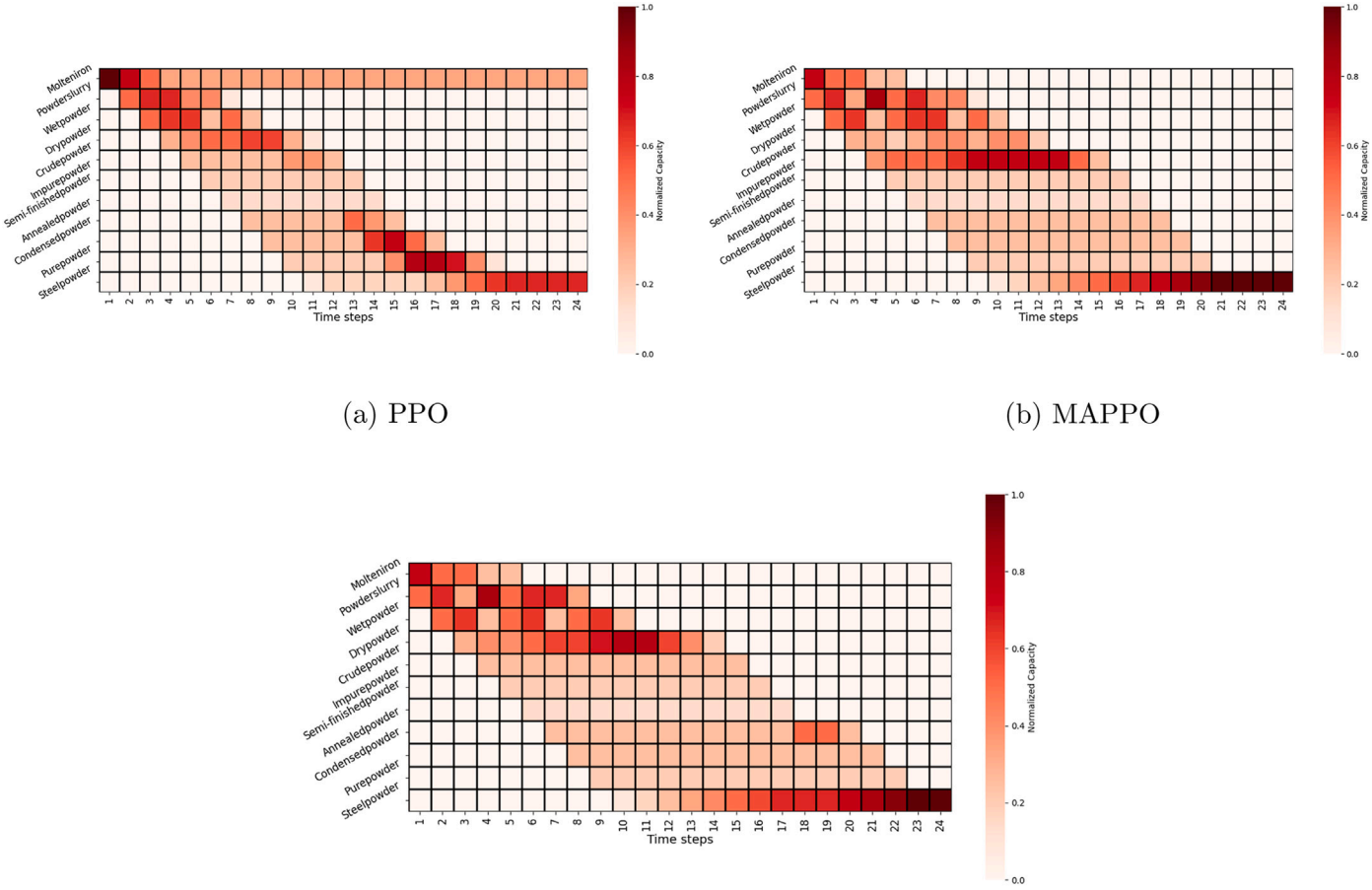
enables quick identification of storage patterns, such as consistently underutilized or overutilized containers, and helps track temporal trends in resource usage. Such a heat map is presented in Fig. 9. It is clear from Fig. 9 that the PPO-based agent does not optimize the production and consumption of resources. The production goal is not yet met, and several bottlenecks are seen during the production process, which hints at uncoordinated load shifting. Conversely, multi-agent methods allow resource storage and utilization to be more coordinated and aligned to aid load shifts. Furthermore, the coordination shown by the attention mechanism is better than that of MAPPO. Such an example is evident in time-steps 18 and 19 when the MAPPO-based scheduling decides to run the equipment despite the high energy pricing period. However, the proposed algorithm schedules the equipment after the high pricing period, utilizing storage efficiently.

Fig. 10 visualizes all shiftable and controllable energy consumption at each time step, offering insights into scheduling behavior. As seen in the image, the equipment responsible for production in the primary process is sequentially scheduled while adjusting loads to achieve an even distribution of shifted loads. Further, the scheduling of auxiliary equipment is more streamlined when electricity prices are low.

Fig. 11 illustrates the electricity demand of the Energy Storage System (ESS) during each stage, where positive values correspond to charging and negative values represent discharging. It is evident that the ESS behavior is mainly achieved during two periods when electricity prices were low; the ESS charged from the primary power grid to store energy. Conversely, during high-price periods, the ESS discharged energy to supply the local industrial facility, thereby reducing energy costs associated with high-price consumption. At each stage, the ESS charged or discharged up to its maximum limit of 50 kWh, constrained by the maximum charge/discharge rate specified in Table A.13. Due to energy losses inherent in the charging and discharging processes, the total amount of discharged electricity was less than the total charged electricity. These results demonstrate that coordination between agents has been fostered by centralized training and decentralized execution of energy devices.

4.3.1. Ablation study: sensitivity analysis of the reward weighting parameter α

To assess the impact of the weighting parameter α in Eq. (19), a sensitivity analysis was conducted by varying α within the range of 0.2 to 1.0. This parameter governs the trade-off between production efficiency and energy cost in the overall reward function. The corresponding reward curves for each α value across training episodes are shown in Fig. 12, illustrating how the reward converges to distinct values depending on the relative emphasis placed on production and cost. The analysis reveals that lower values of α (less than 0.7) prioritize minimizing energy costs, leading the agent to avoid scheduling equipment, which results in no production. As α increases, particularly for values of 0.8 and 0.9, the agent begins to schedule equipment. However, at higher values of α, there is an increasing tendency to favor production throughput. At



(a) PPO

(b) MAPPO

(c) Proposed algorithm

Fig. 9. Container resource monitoring.

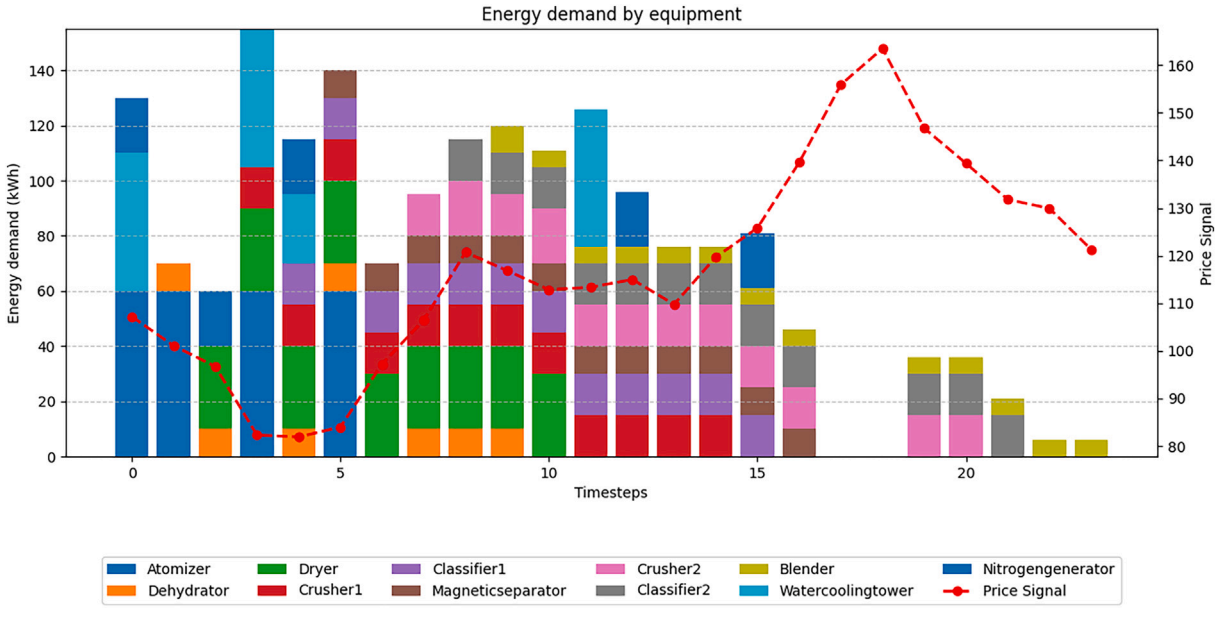


Fig. 10. Comparison of total energy costs.

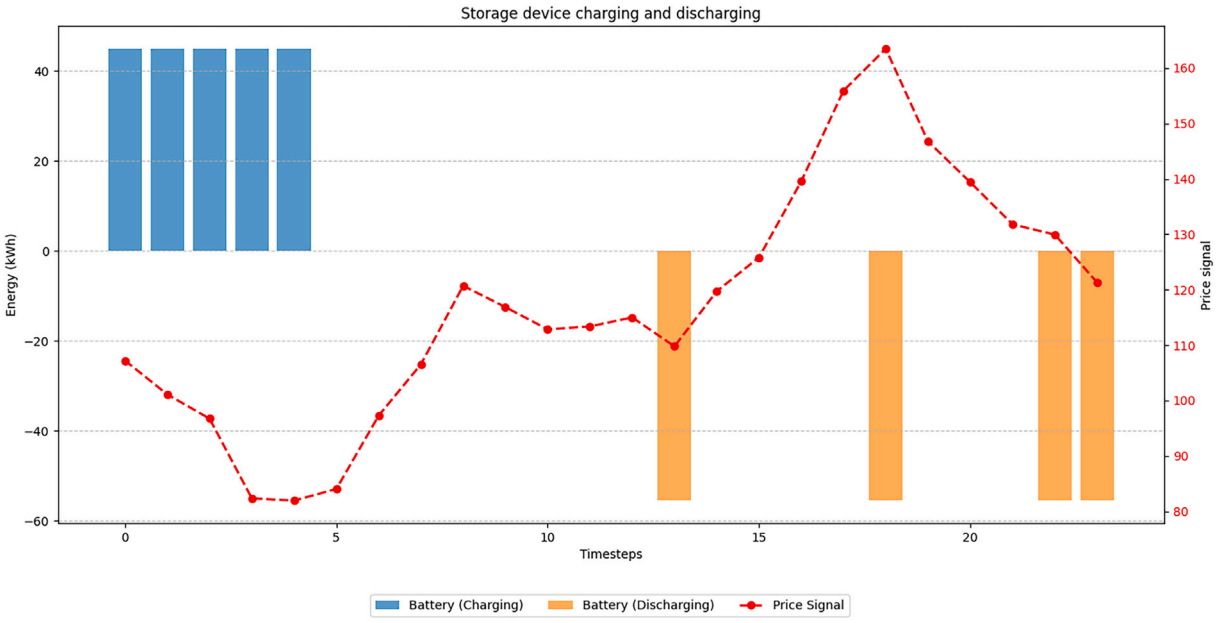


Fig. 11. Comparison of total energy costs.

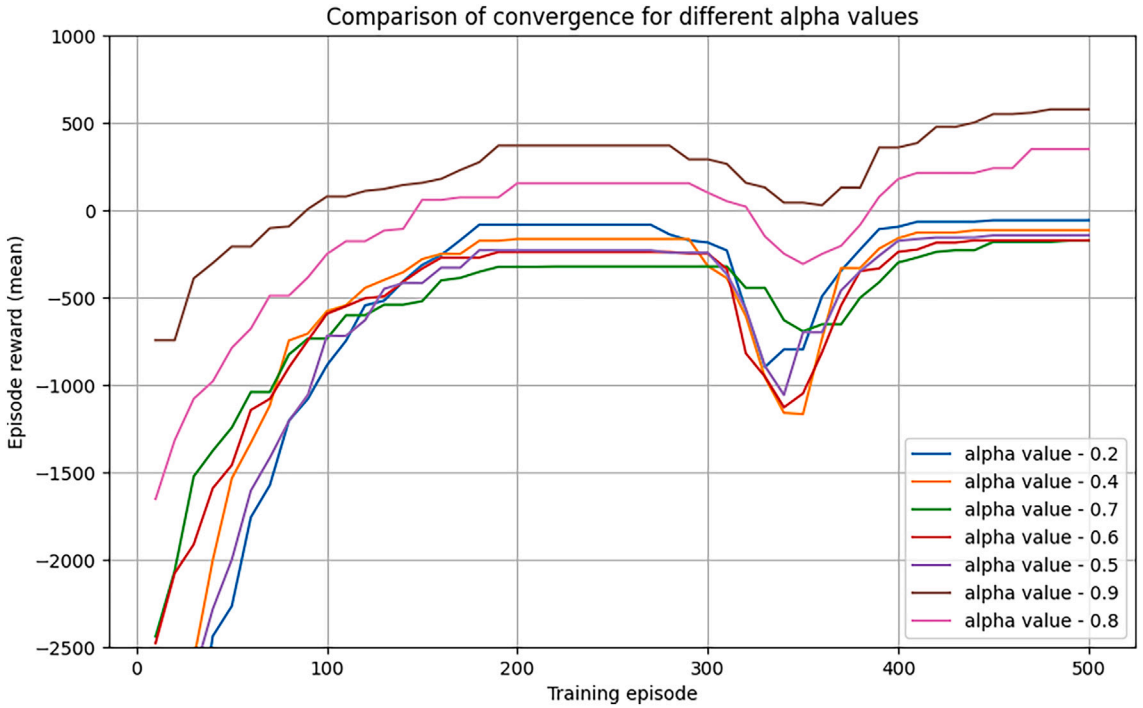
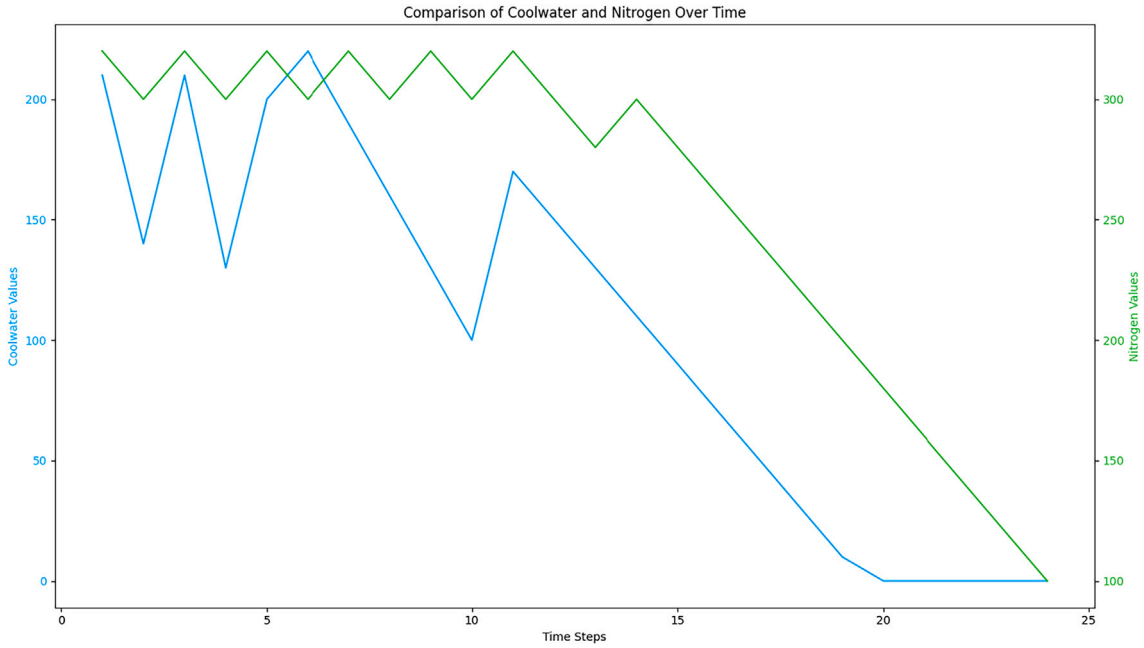


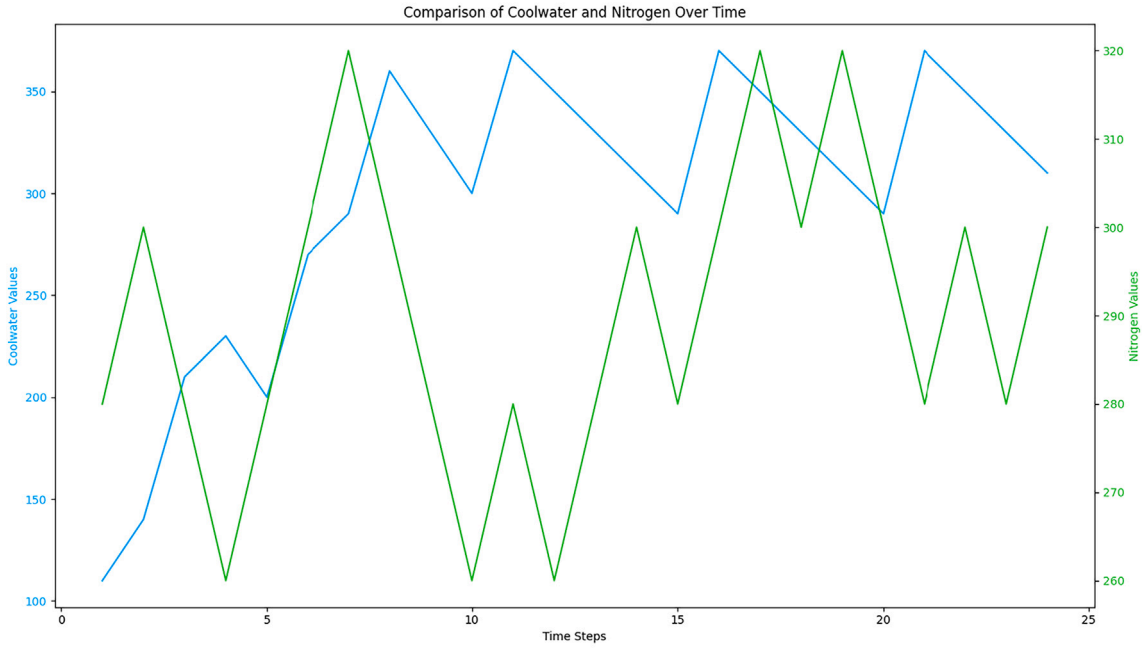
Fig. 12. Comparison of reward evolution for different alpha values in reward function.

$\alpha = 0.9$, the agent's focus on production becomes so pronounced that energy costs are largely disregarded. This effect is further highlighted when the storage of auxiliary resources (i.e. cool water and nitrogen) is compared for α values of 0.8 and 0.9, as shown in Fig. 13. With a higher production reward weight ($\alpha = 0.9$), the agent produces auxiliary resources primarily to maximize production rewards. In contrast, at $\alpha = 0.8$, better coordination is observed, as auxiliary resources are produced only during low electricity price periods and are utilized from storage, demonstrating superior scheduling behavior.

In real-world applications, the relative importance of minimizing energy costs versus maximizing production throughput can vary depending on industrial priorities, market conditions, and operational constraints. Fine-tuning this parameter allows for tailoring the agent's decision-making process to align with specific objectives, such as reducing operational expenses or increasing production output. Therefore, understanding and adjusting such weighting parameters α is essential for achieving optimal performance in dynamic environments, where both cost and production factors must be carefully balanced.



(a) alpha 0.8



(b) alpha 0.9

Fig. 13. Auxilary container resource monitoring.

5. Conclusion and discussion

5.1. Summary of the findings

This paper shows that DR-based scheduling of heavy industries is imperative for achieving decarbonization goals and a sustainable clean-energy ecosystem. To foster practical applications, we demonstrate the design of a POMDP framework for industrial DR and formulate corresponding state, action, and reward functions. The study also presents a framework to develop a MAPPO algorithm that achieves centralized training and decentralized execution of multiple agents. This framework has been shown to improve cooperation among agents, which has practical benefits in industrial settings with heterogeneous energy loads that require cooperation. An attention mechanism is also introduced in the centralized training process to allow the critic architecture to focus on key information, which has been shown to improve the exploration efficiency of MAPPO. Finally, based on accurate data and the parameters of an SPM process, the performance of several DRL algorithms is analyzed, showing that the proposed algorithm is an effective solution for industrial energy management problems. The results demonstrate that utilizing centralized training and a decentralized execution framework

in MARL is a viable and advantageous solution to aid cooperation among loads in complex industrial settings.

5.2. Practical implications

The practical implications of this research are significant for industries aiming to reduce energy consumption while meeting production goals. As shown in this work, one of the main strengths of price-based DR schemes is their simplicity and ease of implementation. Price signals can be easily communicated to industrial loads, effectively controlling energy usage without requiring extensive infrastructure changes. In our framework, using day-ahead pricing allows for optimizing the scheduling of controllable and schedulable loads, ensuring that energy consumption is aligned with dynamic market prices. The framework not only focuses on energy cost reductions but also optimizes energy use to minimize the disruption to production processes.

Additionally, this framework acknowledges the critical aspects of production goals and addresses the need to reduce production bottlenecks while shifting loads. By considering production constraints, such as deadlines and material availability, the algorithm ensures that load shifting does not interfere with key operational processes. Thus, industries can maintain high productivity levels while benefiting from energy cost reductions.

Furthermore, our approach extends the benefits of traditional DR systems by leveraging MARL, which fosters cooperation between heterogeneous industrial loads. This collaboration among loads is particularly beneficial in complex industrial environments where different machines or processes have different energy consumption patterns but must work together for optimal efficiency. Integrating centralized training and decentralized execution promotes efficient resource allocation, ensuring that all agents work towards common goals without needing continuous centralized control.

In real-world applications, this framework can lead to substantial operational savings and more effective load management, contributing to a cleaner and more sustainable energy grid. It allows industries to adopt more dynamic, scalable, and collaborative DR systems, ultimately supporting the transition towards greener, more sustainable industrial operations.

5.3. Limitations and future works

While the proposed framework offers promising solutions, it is essential to acknowledge some limitations and outline avenues for future research. One of the key areas for further investigation is the exploration of various demand response schemes beyond price-based systems. Our study focuses on price signals as the primary mechanism for load scheduling, but in a multi-agent environment, incorporating a variety of DR schemes – such as load curtailment, renewable energy integration, or incentives for flexible demand – could lead to even more sophisticated and practical solutions. Future research could explore applying multiple DR schemes simultaneously, allowing for more complex and dynamic energy management strategies. Another direction of research could be considering the optimization problem as continuous instead of discrete, such that load interruptions are not described as discrete events in time.

Furthermore, while MARL offers clear advantages in terms of cooperation among agents, there remains significant potential for integrating additional types of utilities within the DR framework. For instance, managing heat, hydrogen production, and other utilities alongside electricity could create a more holistic energy management system that better aligns with the broader decarbonization goals and sustainable energy use. Exploring such integrations within the MARL framework could offer valuable insights into how industries can optimize their energy consumption across multiple domains.

While the proposed framework demonstrates faster convergence behavior with the attention based mechanism, it is crucial to address the convergence speed, computational cost, and scalability of the framework in real-world industrial environments. The centralized training paradigm, augmented with an attention mechanism, enhances learning efficiency by prioritizing the most informative features. However, as the scale of the system increases – with more agents and heightened coordination complexity – the convergence time may become a limiting factor. This brings us to optimizing computational cost and stability of the resulting model. The computational cost associated with training the MAPPO algorithm is a critical factor, particularly when considering deployment in industrial environments or scaling the model to larger systems. Our experiments were conducted on a device with moderate computational resources; however, as the number of agents and system complexity grow, computational demands are expected to increase substantially.

To address these aspects, future work should focus on combining the current research with research trends seen in Industrial Internet of Things (IIoT). In modern industries, millions of low-power IoT devices leverage Internet-based technologies alongside local services. While, the concept of IIoT allows real time Machine-to-Machine (M2M) communication, it integrates technologies such as cloud computing, robotics, big data, cyber–physical systems, and artificial intelligence. As such, the research will focus on accelerating convergence through parallelized training strategies and leveraging hardware accelerators such as GPUs and tensor-optimized processing units. Another aspect of future research extension is the exploration of methods to reduce the model's computational footprint. Techniques such as model compression and pruning can make the algorithm suitable for deployment on edge devices, enabling low-latency inference in time-critical control scenarios. While the current framework relies on day-ahead scheduling – where latency is not a primary concern – future extensions involving real-time or near-real-time scheduling (e.g. adjustments in response to low frequency DR signals i.e 15 min price changes) will necessitate faster decision-making. In such cases, addressing computational efficiency while addressing latency becomes crucial to maintaining responsiveness and robustness in uncertain, rapidly changing environments. As such, the deployment of AI-driven strategies for distributing computational resources across industrial infrastructures is an active area of research within the Industrial Internet of Things (IIoT) domain [56]. Finally, another aspect of future research always lies in adaptation of the environment to novel optimization mechanisms and comparative studies with other RL and heuristic-based optimization techniques.

In conclusion, while this study provides a valuable foundation for industrial energy management through MARL and price-based DR, there is much room for further exploration. As the landscape of industrial energy management continues to evolve, future research will play a pivotal role in refining and expanding upon the ideas presented here, ultimately contributing to more sustainable and efficient industrial operations.

CRediT authorship contribution statement

Atit Bashyal: Writing – original draft, Software, Methodology, Formal analysis, Conceptualization. **Tina Boroukhian:** Writing – review & editing, Methodology, Formal analysis. **Pakin Veerachanchai:** Writing – review & editing, Software. **Myanganbayar Naransukh:** Writing – review & editing, Writing – original draft. **Hendro Wicaksono:** Writing – review & editing, Supervision, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT solely in order to improve readability and language. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Steel manufacturing process parameters

See Tables A.6–A.13.

Table A.6
Detailed information on energy demand.

Equipment name	Equipment type	Energy demand (kWh)				
		OP #1	OP #2	OP #3	OP #4	OP #5
Atomizer	SE	0	60	/	/	/
Dehydrator	SE	0	10	/	/	/
Dryer	SE	0	30	/	/	/
Crusher	CE	0	15	20	/	/
Classifier	CE	0	15	25	/	/
Magnetic separator	SE	0	10	/	/	/
Reduction furnace	NSE	75	0	/	/	/
Blender	CE	0	6	10	/	/
Water cooling tower	CE	0	25	50	/	/
Nitrogen generator	CE	0	20	30	40	55

Table A.7
Processing capability of intermediate resources by equipment.

Equipment name	Processing rate (ton/h)		
	OP #1	OP #2	OP #3
Atomizer	0	30	/
Dehydrator	0	20	/
Dryer	0	15	/
Crusher	0	10	15
Classifier	0	10	20
Magnetic separator	0	30	/
Reduction furnace	15	0	/
Blender	0	10	15

Table A.8
Processing capability of cool water by different equipment.

Equipment name	Processing rate (ton/h)		
	OP #1	OP #2	OP #3
Atomizer	0	50	/
Dryer	0	10	/
Reduction furnace	20	/	/
Water cooling tower	0	100	200

Table A.9
Processing capability of nitrogen by different equipment.

Equipment name	Processing rate (ton/h)				
	OP #1	OP #2	OP #3	OP #4	OP #5
Reduction furnace	20	/	/	/	/
Nitrogen generator	0	40	60	80	100

Table A.10
Detailed information on resource storage in the main process.

Resource name	Storage capacity (ton)	Inlet equipment	Outlet equipment
Molten iron	50	/	Atomizer
Powder slurry	60	Atomizer	Dehydrator
Wet powder	40	Dehydrator	Dryer
Dry powder	50	Dryer	Crusher #1
Crude powder	40	Crusher #1	Classifier #1
Impure powder	50	Classifier #1	Magnetic separator
Semi-finished powder	70	Magnetic separator	Reduction furnace
Annealed powder	70	Reduction furnace	Crusher #2
Condensed powder	40	Crusher #2	Classifier #2
Pure powder	50	Classifier #2	Blender
Steel powder	70	Blender	/

Table A.11
Detailed information on cool water storage.

Resource name	Storage capacity (m^3)	Inlet equipment	Outlet equipment
Cool water	900	Water cooling tower	Atomizer Dryer Reduction furnace

Table A.12
Detailed information on Nitrogen storage.

Resource name	Storage capacity (Nm^3)	Inlet equipment	Outlet equipment
Pure nitrogen	700	Nitrogen generator	Reduction furnace

Table A.13
Detailed information on energy storage systems (ESS).

Parameter name	Value
Storage capacity	300 kWh
Discharging efficiency	0.9
Charging efficiency	0.9
Maximum discharging rate	50 kWh
Maximum charging rate	50 kWh

Data availability

Data will be made available on request.

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